



Román Flores Rodríguez

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ABOUT MYSELF

Software Engineer – GIS, Artificial Intelligence & Backend Development COTESA – Spatial Observation Center

Currently working as a Software Engineer developing innovative solutions that combine **Geographic Information Systems (GIS), backend development, microservices, and Artificial Intelligence technologies**.

My role focuses on designing and implementing scalable software solutions for spatial data processing, automation, and intelligent workflows, integrating modern software engineering practices with emerging AI technologies.

Key responsibilities and achievements:

- Develop GIS-based applications and geospatial processing workflows using **Python and spatial data technologies**
- Build automation tools for spatial data analysis, validation, transformation, and quality improvement.
- Design and develop backend services and **microservice architectures using Java and Spring Boot**.
- Integrate **Large Language Models (LLMs) and Generative AI solutions** into real-world applications and business processes.
- Develop data processing pipelines for large-scale geospatial datasets.
- Create and consume REST APIs to connect different software components and services.
- Apply software engineering best practices including clean architecture, code quality, testing, and version control.
- Explore and implement AI-driven solutions to automate processes and improve productivity.

EDUCATION & TRAINING

18/09/2016 - 10/11/2023 - SEVILLA, SPAIN

GRADO EN INGENIERÍA INFORMÁTICA. INGENIERÍA DEL SOFTWARE- UNIVERSIDAD DE SEVILLA

LANGUAGE SKILLS

Mother tongue(s): **ESPAÑOL**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
INGLÉS	C1	B2	C1	B2	B2
FRANCÉS	B1	B1	B1	B1	B1

SKILLS

BackEnd

Django (basic) | AI-Gen | Github Copilot | Generative-AI | Java | SPRING BOOT | WSL2 | WSO2 | Python | Git | Selenium in Python | Basic QGIS skills | geographic information systems

FrontEnd

HTML (Basics) | Html / CSS / JavaScript

AI's

chatgpt | Tensorflow Keras | Copilot | dall-e | RAG | CSS | LLM | Python AI

DDBB

MySQL | PostgreSQL + SQLite + SQLServer + MySQL | Oracle Warehouse Builder | PostgreSQL, SQLite, SQLServer y MySQL | SQL

● WORK EXPERIENCE

02/11/2025 - CURRENT - SEVILLA, SPAIN

SOFTWARE ENGINEER – GIS, MICROSERVICES & GENERATIVE COTESA – SPATIAL OBSERVATION CENTER

Software Engineer specialized in Geographic Information Systems (GIS), backend development, microservices architectures, and Generative Artificial Intelligence solutions applied to spatial data and enterprise systems.

Worked on the design and implementation of scalable geospatial platforms by combining GIS technologies, software engineering best practices, and AI-driven architectures to process, analyze, and automate large-scale spatial data workflows.

Key responsibilities:

Design and development of GIS-based applications and geospatial data processing systems using Python and spatial data libraries.

Development of automation pipelines for geospatial data validation, transformation, quality control, and analysis.

Design and implementation of backend services and microservices architectures using Java and Spring Boot.

Development and integration of RESTful APIs for interoperability between geospatial systems and enterprise platforms.

Definition and implementation of software architectures for scalable and distributed systems.

Design and integration of Generative AI solutions applied to geospatial analysis, automation, and decision-support systems.

Research and implementation of RAG (Retrieval-Augmented Generation) architectures applied to structured and unstructured spatial data.

Exploration and integration of Large Language Models (LLMs) into GIS workflows and enterprise software solutions.

Development of data processing pipelines for large-scale geospatial datasets and spatial intelligence systems.

Application of software engineering best practices including clean architecture, testing, CI/CD, and version control workflows.

01/06/2025 - CURRENT - SEVILLA

SOFTWARE ENGINEER – MICROSERVICES ARCHITECTURE & ARTIFICIAL INTELLIGENCE | WSO2 API MANAGEMENT NTT DATA

Software Engineer specialized in designing and developing **enterprise architectures based on Java, Python, Microservices, and Artificial Intelligence technologies.**

Key responsibilities:

- Design and implementation of **production-grade software architectures** using Java, Python, and AI-based solutions.

- Design, development, and governance of enterprise APIs using **WSO2 API Manager**, following API-first principles and OpenAPI standards.
- Development and integration of solutions based on the **WSO2 technology stack**.
- Direct collaboration with clients, gathering requirements and defining technical solutions aligned with business needs.
- Team coordination, technical planning, and support in project management activities.
- Development and integration of **Generative AI solutions** and intelligent automation workflows.

18/10/2023 - 01/06/2025 - SEVILLA, SPAIN

JUNIOR SOFTWARE ENGINEER – SOFTWARE ARCHITECTURE & GENERATIVE AI NTT DATA

Junior Software Engineer specialized in **software architecture, microservices development, and Generative Artificial Intelligence solutions**.

Worked on the design and implementation of modern software architectures, combining backend development practices with emerging AI technologies to build scalable and intelligent solutions.

Key responsibilities:

- Design and development of software architectures based on **Java Spring Boot (Spring 2 and Spring 3)**
- Development and integration of **microservices architectures**, including service communication and modular system design.
- Containerization and deployment of applications using **Docker**, improving scalability and consistency across environments.
- Design and implementation of **Generative AI solutions** to support software development and business processes.
- Definition of **RAG (Retrieval-Augmented Generation) architectures** and integration of **Large Language Models (LLMs)** to software solutions.
- Creation of training programs and technical documentation focused on **Generative AI adoption and GitHub Copilot best practices**
- Research and evaluation of AI-assisted development approaches to improve engineering productivity.